Auteur: Mathijs Pittoors
Studierichting: Informatica
Oefeningenreeks 4A nr. 27.17

$$\int \frac{x^4+x^3+3x^2+x+1}{x^3+1} dx$$

Euclidische deling:

$$\begin{array}{ccccc|cc}
x^4 & +x^3 & +3x^2 & +x & +1 & x^3 & +1 \\ \cline{6-7}
-x^4 & & & -x & & x & +1 \\ \cline{1-5}
 & x^3 & 3x^2 & & +1 & & \\
 & -x^3 & & & -1 & & \\ \cline{1-5}
 & & 3x^2 & & & &
\end{array}$$

$$\Rightarrow \int x+1dx + \int \frac{3x^2dx}{x^3+1} = \frac{x^2}{2} + x + \int \frac{3x^2dx}{x^3+1}$$

Neem $t = x^3+1$ dan $dt = 3x^2dx$ dus $dx = \dfrac{dt}{3x^2}$

dan $\dfrac{x^2}{2} + x + \displaystyle\int \frac{dt}{t} = \frac{x^2}{2} + x + \ln|x^3+1| + c$ met $c \in \mathbb{R}$

References